RCS Solver Validation Report

Date: 2025-11-22

Project: RCS Cone Validation
Target: 1 m cone, 0.5 m base, 0.1 m tip radius

CPU Solver: MoM/PO - CPU Baseline
CPU Hardware: Xeon Gold (example)

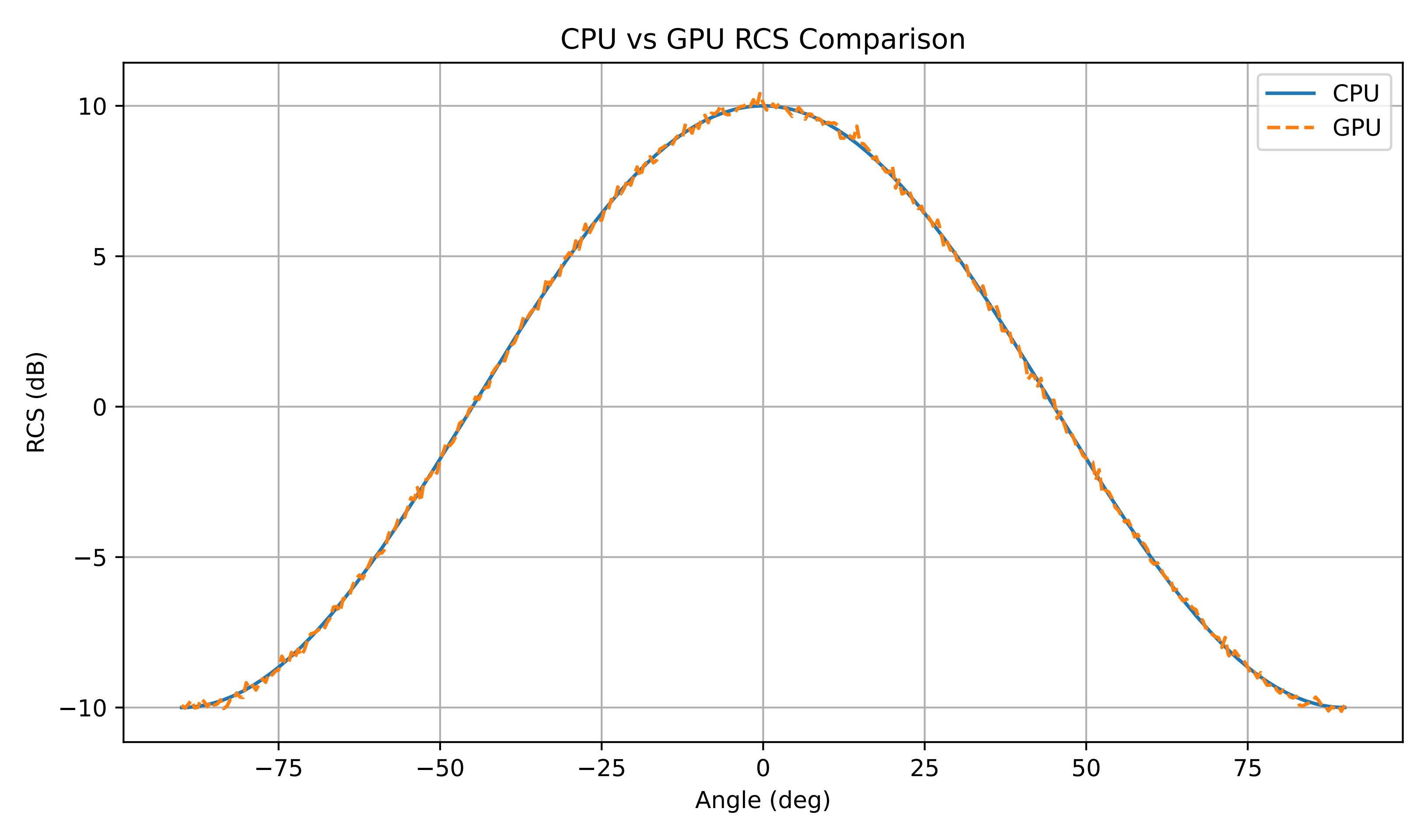
GPU Solver: MoM/PO - GPU Accelerated
GPU Hardware: NVIDIA A100 (example)

Frequency: 10.0 GHz
Polarization: HH

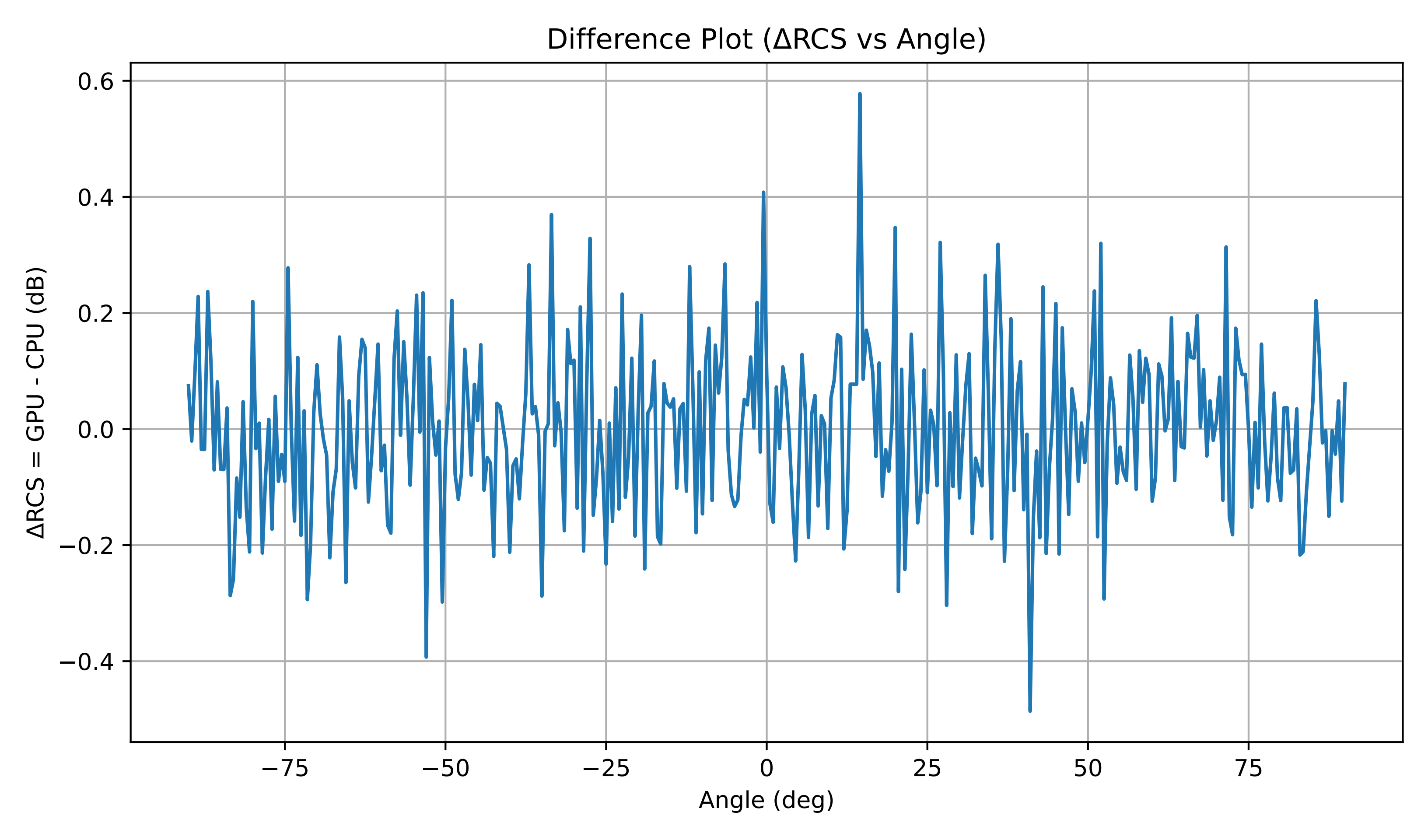
Summary of Key Metrics:

```
  Max |ΔRCS|          : 0.578 dB
  Mean |ΔRCS|         : 0.112 dB
  95th pct |ΔRCS|     : 0.283 dB
  RMS diff (linear)   : 1.503e-01
  NRMSE (linear)      : 1.503e-02
  Pattern correlation : 0.99980
```

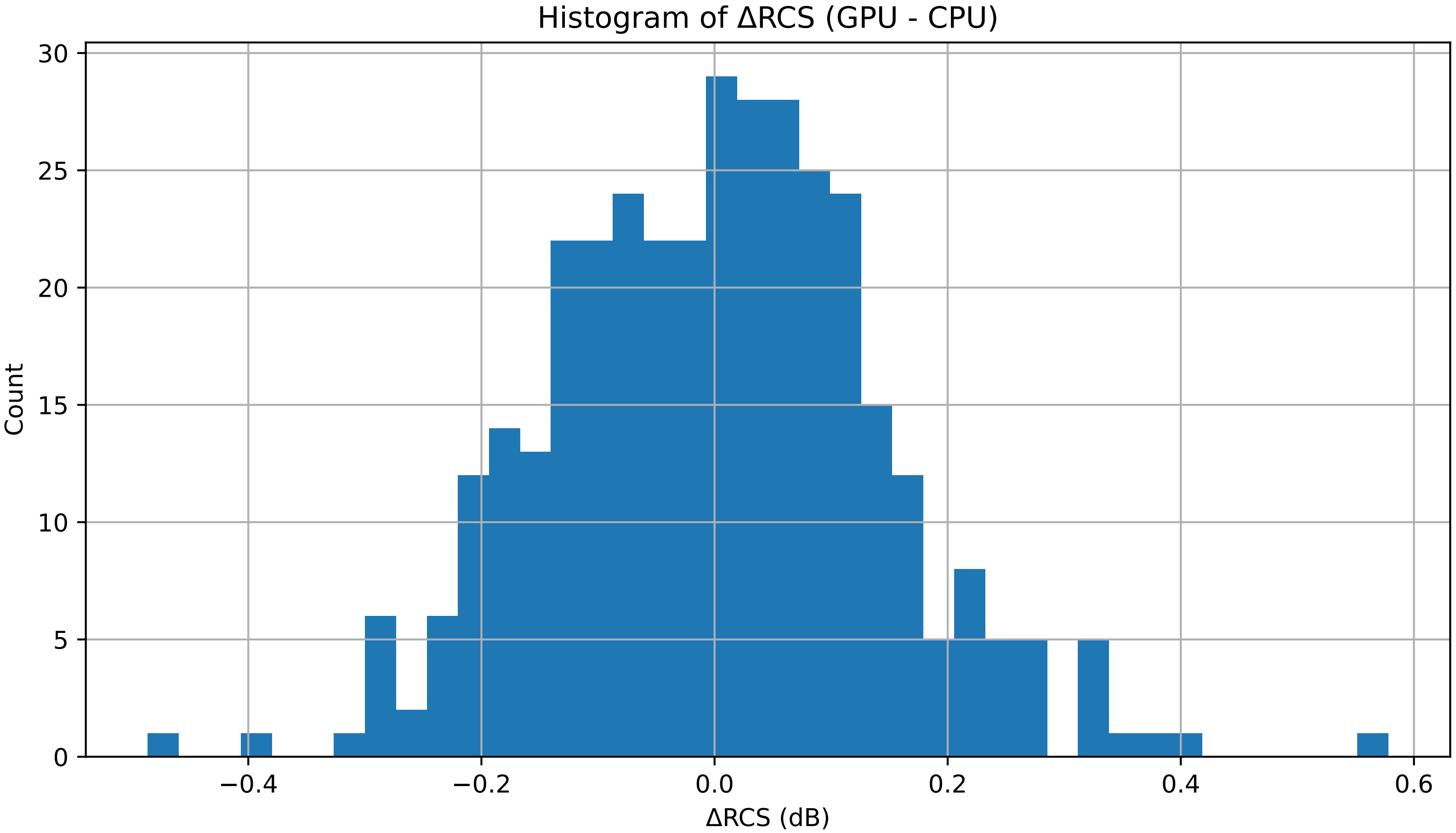
```
Detailed Accuracy Metrics

Amplitude (dB-domain):
  Max |ΔRCS| (dB)        : 0.577910
  Mean |ΔRCS| (dB)       : 0.112428
  95th percentile |ΔRCS| : 0.282928

Linear-domain metrics:
  RMS difference (σ_lin) : 1.503064e-01
  NRMSE (σ_lin)          : 1.503064e-02

Pattern correlation (dB-domain):
  Corr. coefficient      : 0.999800

Interpretation guidelines (typical):
  Max |ΔRCS| < 0.2 dB  → Excellent match
  Max |ΔRCS| < 0.5 dB  → Good/acceptable
  Max |ΔRCS| > 1.0 dB  → Investigate (precision/tolerance issues)

NRMSE (linear domain):
  < 1%   → Very high fidelity
  1–5%   → Acceptable for most PO/MoM RCS work
  > 10%  → Potential numerical or modeling issue
```



CPU vs GPU RCS Comparison
CPU
GPU
RCS (dB)
Angle (deg)
10
5
0
−5
−10
−75
−50
−25
0
25
50
75



Difference Plot (ΔRCS vs Angle)
0.6
0.4
0.2
0.0
−0.2
−0.4
ΔRCS = GPU - CPU (dB)
−75
−50
−25
0
25
50
75
Angle (deg)


Histogram of ΔRCS (GPU - CPU)
Count
ΔRCS (dB)
0
5
10
15
20
25
30
−0.4
−0.2
0.0
0.2
0.4
0.6